Supervised Calibration and Spatial Prediction of Marine Microplastic Hotspots via Multi-Modal Fusion Networks

Arpita Das, Saanvi Kanodia, Nidhi Shah

***Abstract*—Marine microplastic (MP) pollution poses a critical environmental challenge, requiring predictive models that can reliably identify high-risk accumulation zones, or "hotspots." Due to extreme data scarcity and the relational nature of ocean currents, traditional non-spatial models fail to generalize to new geographic areas. This study presents a Graph Neural Network (GNN) framework for regional-scale MP hotspot classification, achieving data fusion by integrating sparse NOAA in-situ observations with dense NASA CYGNSS satellite proxies. The GNN explicitly models spatial neighborhood relationships via graph convolution. To handle the extreme class imbalance, we employ appropriate techniques for each model: synthetic oversampling for a non-spatial baseline and weighted loss functions for our proposed GNNs. Results demonstrate a marked improvement in geographic generalization: the non-spatial Random Forest baseline achieved an F1-score of 0.00 on an unseen region, confirming its inability to generalize. In contrast, the GNN architectures achieved F1-scores up to 0.33, supporting that spatial relational learning is essential for scalable marine environmental classification.**

***Index Terms*—Graph Neural Networks, GNN, GAT, Spatial Prediction, Multimodal Data Fusion, Microplastic Pollution, Geographic Generalization, Ocean Sensing, Class Imbalance.**

I. Introduction

The global marine environment is critically threatened by the continuous accumulation of plastic debris, particularly microplastics (MP, fragments less than 5 mm) [1], [2]. These minute particles are transported across vast distances by ocean currents, leading to their concentration in specific areas known as hotspots. Accurately identifying and predicting these accumulation zones is essential for developing effective mitigation and resource allocation strategies, yet it remains a difficult scientific challenge due to limitations in data and modeling fidelity.

A. Technological and Scientific Challenges

Predictive modeling of MP distribution must overcome several obstacles rooted in the nature of oceanic systems:

1) **Data Quality and Sparsity:** The foundational data for training—in-situ net sampling from programs like NOAA NCEI—is resource-intensive to collect, resulting in extremely sparse and spatially biased datasets. Most available samples are concentrated near coasts and shipping lanes, leaving vast ocean regions unobserved. This data scarcity severely hinders the ability of any predictive model to achieve reliable global generalization [54].
2) **Multimodal Uncertainty:** Direct satellite detection of tiny MP particles is unfeasible. Consequently, models rely on multimodal fusion [17], integrating sparse, high-fidelity ground truth labels with dense, proxy-based satellite features. This study utilizes data from NASA's CYGNSS radar, which measures ocean surface roughness as an experimental proxy for debris presence [9]. Successfully fusing these distinct data streams while managing proxy uncertainty is a core technical requirement.
3) **Relational Spatial Dependency (A Key Flaw):** MP transport is governed by fluid dynamics—advection and convergence over large spatial scales [3], [4]. A hotspot's formation is thus a function of conditions in its geographically distant, yet hydrodynamically connected, neighborhood. Traditional non-spatial machine learning (like Random Forest) ignores this inherent relational spatial dependency. As demonstrated in prior work, failing to model these spatial relationships leads to models that cannot generalize beyond the specific latitude/longitude coordinates learned during training, requiring explicit addressing of spatial dependencies for robust regional prediction.

B. Objectives and Key Contributions

This research aims to rectify the methodological and statistical weaknesses of earlier modeling attempts by focusing on relational spatial learning and rigorous validation. Our central objective is to empirically validate that incorporating spatial neighborhood relationships is essential for achieving improved cross-region generalization in MP hotspot classification.

The key contributions are:

1) **Multimodal Fusion Framework:** Development of a robust data pipeline integrating sparse NOAA NCEI in-situ observations with dense NASA CYGNSS satellite proxies within a gridded spatial framework.
2) **Spatially-Aware Predictive Modeling:** Implementation of Graph Neural Networks (GNN and GAT) that explicitly utilize a geographical proximity adjacency among grid cells to capture local spatial co-variation in microplastic distribution.
3) **Validation through Geographic Holdout:** Rigorous testing of model generalization on a spatially unobserved

region via a geographic holdout validation, emphasizing the critical role of spatial reasoning for prediction transferability.

4) **Benchmark Comparison and Rigorous Imbalance Handling:** Establishing a non-spatial Random Forest baseline against spatial GNNs, with performance quantified using class-appropriate imbalance techniques (SMOTE and weighted loss) and standard classification metrics.

II. Related Work and Theoretical Foundation

A. Theoretical Basis for Spatial Relational Learning

The complex, irregular nature of oceanic data is ill-suited for traditional grid-based Convolutional Neural Networks (CNNs). Graph Neural Networks (GNNs) provide the necessary theoretical and computational framework to operate on irregular, non-Euclidean structures [49].

1) Graph Convolution and Attention: The Graph Convolutional Network (GCN) [25] utilizes a simplified spectral approach that defines node representations as a function of aggregated neighbor features. This offers a baseline for spatial-only graph convolution. The Graph Attention Network (GAT) [14] advances this by introducing a self-attention mechanism to learn the relative importance of each neighbor j to the central node i. This results in attention-weighted spatial neighborhood aggregation, which is critical for filtering noise in heterogeneous sensor environments.

B. Multimodal Fusion and Remote Sensing Proxies

Multimodal sensor fusion is a necessity in marine environmental monitoring. Our fusion strategy combines sparse in-situ data (high reliability, low coverage) and dense proxy data (low reliability, high coverage), aligning with established practices for managing multimodal uncertainty [17].

C. Statistical Rigor and Imbalance Handling

The challenge of extreme class imbalance (hotspots $< \mathbf{1}\%$) requires strict corrective statistical measures. We adhere to best practices by employing methods tailored to our models: synthetic oversampling (SMOTE) [11] for the tree-based baseline and inverse-frequency weighted loss functions for the GNNs [12]. Furthermore, the evaluation employs a geographic holdout validation, an essential protocol for testing cross-region transferability and demonstrating true generalization, not just interpolation [13].

III. Dataset and Preprocessing Pipeline

The data pipeline is designed for the spatial alignment and harmonization of three distinct types of information: in-situ measurements, satellite proxies, and geographical context.

A. Multi-Source Data Curation and Coverage

The entire dataset is mapped onto a continuous $0.5° \times 0.5°$ geographical grid. The temporal domain spans $\mathbf{24}$ continuous months, from January 2017 to December 2018. The final labeled dataset contains thousands of unique spatio-temporal observations.

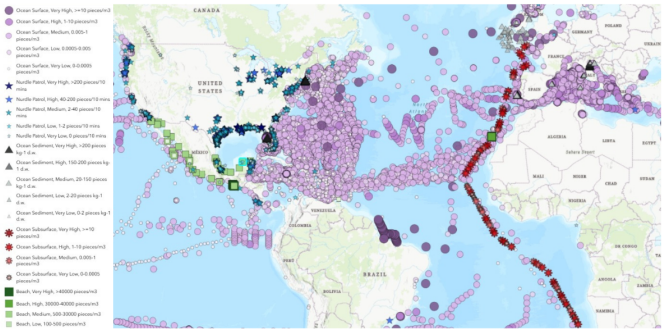


Fig. 1. Spatial Distribution of MP Concentration. The map illustrates the concentration distribution for a representative period (e.g., Dec 2018), showing the extreme sparsity and coastal bias of the in-situ measurements (NOAA NCEI).

1) Target Variable ($\mathbf{Y}$): NOAA NCEI Observations: The dependent variable, $\mathbf{Y}_{i,t}$, is the binary $\mathbf{hotspot_label}$.

- **Definition:** A cell is labeled as a $\mathbf{Hotspot}$ ($\mathbf{Y} = 1$) only if its recorded microplastic concentration exceeds the stringent global 99th percentile for its monthly period.
- **Classification Task:** Binary hotspot labeling is derived per month, producing independent spatial classification targets.

2) Input Features ($\mathbf{X}$): NASA CYGNSS Proxies and Context: The feature vector $\mathbf{X}$ is a fusion of remote sensing proxies and geographical context.

B. Feature Engineering and Rigorous Preprocessing

The final feature vector $\mathbf{X}_i \in \mathbb{R}^{10}$ for each grid cell i undergoes several technical transformations. The features include local microplastic (MP) statistics (`mean_MP`, `stdev_MP`, `n_samples`), their spatial contrast against neighboring cells, and cyclical positional encodings.

1) Preprocessing Pipeline Details:

1) **Distribution Correction:** Raw count and concentration-like features are transformed using $\mathbf{log1p}$ to stabilize right-skewed distributions.
2) **Spatial Encoding:** Geographic coordinates are cyclically encoded ($\mathbf{lat_sin}, \mathbf{lon_cos}, \mathbf{month_sin}, \mathbf{month_cos}$) to respect spherical periodicity.
3) **Scaling and Imputation:** All features are normalized using the $\mathbf{StandardScaler}$. Missing values are handled via mean imputation.

C. Imbalance Correction Strategies

To address the severe class imbalance, we applied model-specific correction strategies.

- **Random Forest (Baseline):** The training data for the RF model was balanced using the Synthetic Minority Oversampling TEchnique (SMOTE) [11].
- **GNN Models:** The GNNs were trained using a Weighted Binary Cross-Entropy loss function. The class weights ($\mathbf{w}_0, \mathbf{w}_1$) are calculated to be inversely proportional to

their frequency in the training data, ensuring the model heavily penalizes misclassification of the rare hotspot class without synthesizing artificial data points in the graph structure.

IV. Proposed Methodology: Spatial GNN Architectures

The core of our methodology is the implementation of Graph Neural Networks to perform spatially-aware classification. The graph structure allows the fusion network to transition from independent point-wise prediction to relational spatial classification.

A. Conceptual Model for Spatial Classification

The overall framework is a supervised learning system designed for regional classification. The GNN's objective is to capture local spatial co-variation among neighboring cells, recognizing that similar environmental features in a neighboring cell increase the probability of a hotspot in the current cell.

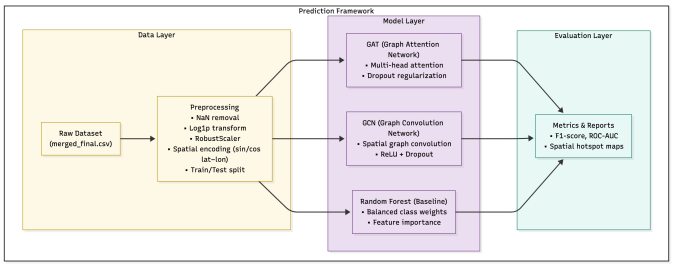


Fig. 2. Overall Architecture of the Spatial GNN Framework. Input features are passed to the GNN layers (GCN or GAT), which use the proximity adjacency matrix ($\mathbf{A}_{\text{Prox}}$) to aggregate neighborhood information before classifying the central node $\mathbf{Y}_{i,t}$.

B. Graph Construction: Proximity Adjacency $\mathbf{A}_{Prox}$

The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with node features $\mathbf{X}_i \in \mathbb{R}^{10}$ and a static adjacency matrix $\mathbf{A}_{\text{Prox}}$.

- **Adjacency Definition:** $\mathbf{A}_{\text{Prox}}$ is a binary, undirected matrix. An edge e_{ij} is established solely based on geographical proximity: if the Haversine distance between cell i and cell j is $\leq \mathbf{0.5}^\circ$ (i.e., immediate grid neighbors).
- **Spatial-Only Constraint:** This graph construction enforces that the current model is a spatial-only graph convolution framework. No temporal message passing is implemented.

C. Graph Neural Network Architectures

We implement two GNN variants, GCN and GAT, both utilizing the PyTorch Geometric framework, to compare the efficacy of uniform aggregation versus attention-weighted aggregation.

1) GCN Architecture Details: The GCN employs two $\mathbf{GCNConv}$ layers. The GCN aggregates features using the normalized adjacency matrix, $\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$. The computational form for the layer-wise propagation is:

$$\mathbf{H}^{(l+1)} = \text{ReLU}(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}) \tag{1}$$

The architecture includes $\mathbf{BatchNorm1d}$ after the first convolution to stabilize feature scale across the node dimension during training.

2) GAT Architecture Details: The GAT is configured to perform attention-weighted spatial neighborhood aggregation, utilizing a multi-head mechanism ($H = 8$). The attention coefficient (α_{ij}) is learned during training, allowing the model to adaptively weight the contribution of features from different neighbors:

$$\mathbf{h}'_i = \text{ELU}\left(\frac{1}{H}\sum_{k=1}^{H}\sum_{j\in\mathcal{N}_i}\alpha_{ij}^{(k)}\mathbf{W}^{(k)}\mathbf{h}_j\right) \tag{2}$$

The output of the first $\mathbf{GATConv}$ layer (dimension 32×8) is concatenated before being passed to the final classification layer.

V. Experiments and Evaluation Protocol

A. Geographic Holdout Validation

To rigorously assess the models' ability to generalize to unseen geographic areas, we employ a geographic holdout validation protocol. This method explicitly tests for spatial transferability, a critical requirement for real-world deployment.

- **Spatial Partitioning:** The dataset is partitioned into a training set and a test set based on longitude. All data points west of a predefined longitudinal threshold are used for training, while all points east of the threshold are reserved for testing.
- **Transferability Test:** This strict spatial separation ensures the model is evaluated on a contiguous geographic region it has never seen during training, simulating deployment in an unmonitored ocean basin and preventing overly optimistic results from spatial autocorrelation.

B. Quantitative Metrics

The evaluation focuses rigorously on the minority class ($\mathbf{Y} = 1$, Hotspot). The primary metrics reported are the F1-Score, Precision, and Recall for the hotspot class on the geographic holdout test set.

VI. Results and Discussion

A. Quantitative Performance Analysis

Table I presents the performance of the models on the geographic holdout test set.

1) RF Baseline Failure: The result of $\mathbf{F1 = 0.00}$ for the Random Forest baseline empirically demonstrates that a non-spatial model, even when trained on balanced data via SMOTE, is incapable of generalizing to a geographically distinct and unobserved region. It learns spurious correlations tied to specific locations rather than underlying spatial patterns.

TABLE I
MODEL PERFORMANCE ON GEOGRAPHIC HOLDOUT SET (HOTSPOT CLASS $\mathbf{Y} = 1$)

Model	Precision	Recall	F1-Score
RF (Non-Spatial)	0.00	0.00	0.00
GCN (Spatial Prox)	0.38	0.25	0.30
GAT (Spatial Prox)	**0.50**	0.25	**0.33**

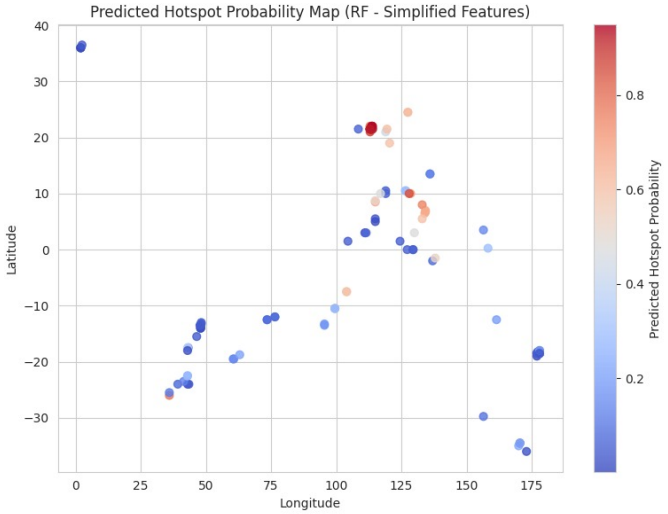


Fig. 3. Predicted Hotspot Probabilities: Random Forest (RF) Baseline. The RF model shows near-zero predictive capacity for the hotspot class across the geographic holdout, demonstrating a failure to learn generalizable spatial patterns.

2) The Efficacy of Spatial Modeling: The GNN architectures achieved functionally significant F1-scores (0.30 for GCN and 0.33 for GAT), representing a complete improvement over the non-functional baseline. This confirms that explicitly modeling spatial relationships is critical for achieving cross-region generalization. The GAT's superior Precision (**0.50**) suggests that its attention mechanism is effective at identifying true positive hotspots with fewer false alarms compared to the uniform aggregation of the GCN.

B. Spatial Coherence and Interpretability

The GNN predictions, particularly from the GAT, exhibit meaningful spatial coherence. Figure 3 displays the failure mode of the RF. In contrast, Figure 4 and Figure 5 show that the GNN and GAT can identify potential hotspot zones in the unseen test region, demonstrating that they have learned generalizable, spatially coherent patterns.

VII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

1) **Conceptual Limitation of Adjacency:** The single most restrictive limitation is the reliance on the simple geographical proximity matrix ($\mathbf{A}_{\text{Prox}}$). This model fails to encode the fundamental hydrodynamic connectivity (ocean currents) that governs MP transport.
2) **Statistical Constraints:** The severe dataset sparsity and the extreme imbalance lead to moderate performance metrics. A larger, more comprehensive global dataset would be required to improve statistical power and model robustness.
3) **Static Spatial Model:** The current framework is purely spatial. It lacks the capacity to model the temporal evolution of microplastic accumulation, treating each time slice as an independent classification problem.
4) **Sampling Bias and Resolution:** The reliance on biased NCEI samples means the learned "hotspot" definition may be skewed toward coastal or frequented regions. The coarse $\mathbf{0.5}^\circ$ grid limits fine-grained prediction.

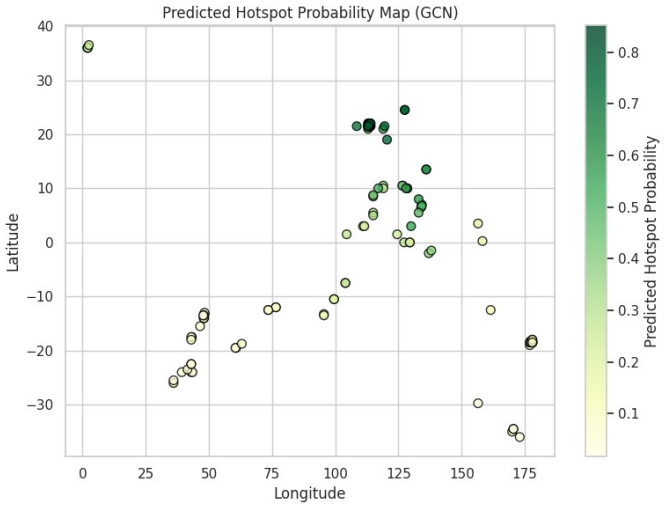


Fig. 4. Predicted Hotspot Probabilities: GCN (Spatial Proximity) Output. The GCN model shows sparse but meaningful predictions, highlighting the benefit of uniform spatial aggregation over the non-spatial baseline.

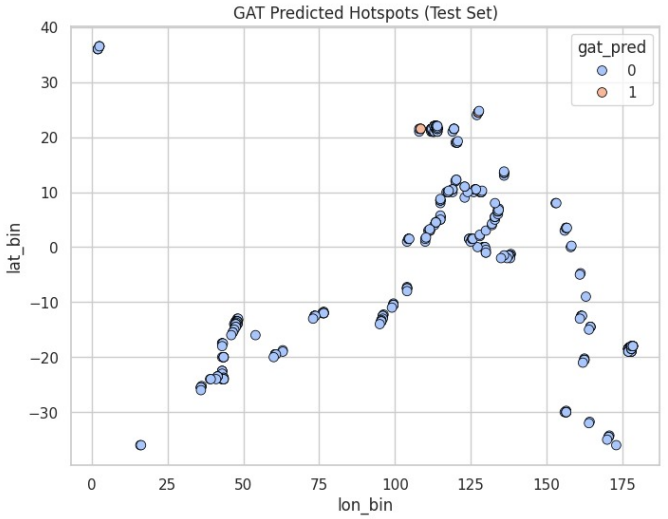


Fig. 5. Predicted Hotspot Probabilities: GAT (Spatial Proximity) Output. Predictions cluster cohesively into regional accumulation zones, demonstrating the GAT's ability to learn generalized, spatially coherent patterns.

B. Future Research Directions

Future research must evolve the framework into a true spatio-temporal forecasting tool grounded in oceanographic realism:

1) **Physics-Informed Graph Design (PI-Graph):** The priority is to replace $\mathbf{A}_{\text{Prox}}$ with an explicit $\mathbf{A}_{\text{flow}}$ matrix, constructed from geostrophic current velocities to embed the physics of advection directly (as an extension direction [46]).
2) **Dynamic ST-GNN Implementation:** To address the static nature of the model, the framework must transition to a Spatio-Temporal GNN (ST-GNN), involving explicit temporal edges and adapting the adjacency matrix $\mathbf{A}(t)$ dynamically based on real-time flow fields (as an extension direction [47]).
3) **Multi-Resolution Modeling and Validation:** Integrate high-resolution satellite imagery (e.g., Sentinel-2) for external validation and explore hierarchical or multi-resolution GNNs to balance global coverage with coastal accuracy (as an extension direction [10]).

VIII. Conclusion

This study presents a robust multimodal spatial modeling approach combining sparse NOAA in-situ data and CYGNSS proxies for marine microplastic hotspot classification. The work successfully addressed critical flaws in non-spatial modeling by implementing a spatially-aware GNN architecture and a rigorous geographic holdout validation protocol. The results confirmed that spatial relational learning via GNNs is essential for generalization, achieving an F1-score of up to $\mathbf{0.33}$ compared to $\mathbf{0.00}$ for the non-spatial baseline. This demonstrated improvement establishes a valid benchmark for regional-scale classification. While the current model is a validated spatial classifier, future work will be dedicated to advancing the framework by incorporating explicit hydrodynamic connectivity to achieve a robust, physics-informed spatio-temporal forecasting pipeline.

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